

Healthcare Appointment Capacity & Queue Management Platform

Implementation Timetable — 20 to 30 September 2026

Goal: Reach a demo-ready, integrated version by 30 September. No AI/ML. Core functionality first; differentiators second.

Daily rule: Finish and test the day's target before moving to optional work. Keep the final two days for integration, bug fixing and presentation preparation.

20 Sep (Sun) — Project setup + database foundation

- Freeze scope and screens.
- Create Spring Boot project and React project.
- Connect Spring Boot to MySQL.
- Configure Hibernate/JPA and JDBC.
- Create database and core tables: users, roles, patients, departments, doctors.
- Test one entity end-to-end with a REST endpoint.

21 Sep (Mon) — Authentication + base CRUD

- Implement users/roles and login.
- Configure Spring Security and password hashing.
- Create patient CRUD.
- Create department CRUD.
- Create doctor CRUD.
- Connect React login and basic management screens.

22 Sep (Tue) — Availability management

- Implement doctor availability.
- Implement working days and consultation hours.
- Implement breaks.
- Implement doctor leave.
- Create backend validation for unavailable periods.
- Build React availability UI.

23 Sep (Wed) — Appointment engine

- Create appointment type/duration.
- Implement appointment booking.
- Validate doctor availability.
- Prevent double booking.
- Validate breaks, leave and appointment duration.
- Implement appointment cancellation and status.
- Connect booking UI to backend.

24 Sep (Thu) — Rescheduling + history

- Implement rescheduling.
- Validate the new slot with the same booking rules.
- Create appointment history.
- Track completed, cancelled, missed and rescheduled states.
- Test edge cases: overlapping appointments, leave, break and invalid duration.

25 Sep (Fri) — Queue management

- Create queue_entries.
- Generate token/queue position.
- Implement queue status changes.
- Implement priority categories/rules.
- Update queue after completion/cancellation.
- Create doctor queue screen and receptionist queue screen.

26 Sep (Sat) — Standout Feature 1: Smart slots

- Build rule-based recommended-slot service.
- Input: doctor/department, date, appointment type.
- Check availability, breaks, leave and existing bookings.
- Return suitable slots.
- Display recommended slots in React.
- Test at least 10 booking scenarios.

27 Sep (Sun) — Standout Feature 2: Waiting time + utilisation

- Calculate estimated waiting time from queue and consultation duration.
- Add doctor utilisation calculation.
- Add department workload calculation.
- Create dashboard cards/charts.
- Clearly label waiting time as an estimate.

28 Sep (Mon) — Cancellation recovery + audit

- Make released appointment capacity available again.
- Add eligible-slot recovery flow.
- Implement audit_logs.
- Record appointment and queue changes with user and timestamp.
- Add audit-log screen.
- Test role permissions.

29 Sep (Tue) — Full integration + testing

- Connect all React screens to real backend APIs.
- Test complete patient → appointment → queue → dashboard flow.
- Test all roles.

- Test invalid inputs and database constraints.
- Fix UI/API/database issues.
- Add what-if simulator only if the core system is stable.

30 Sep (Wed) — Finalisation + demo readiness

- Freeze new features.
- Fix remaining critical bugs.
- Clean UI and error messages.
- Verify database with realistic sample data.
- Prepare project report/screenshots.
- Prepare 5–7 minute demo flow.
- Prepare architecture, database and feature explanation for viva.
- Create final backup/version of source code and database.

Daily Work Pattern

Block	Suggested activity
Block 1	Backend/database implementation
Block 2	Frontend integration
Block 3	Testing and bug fixing
Final 20–30 min	Commit/backup, note completed work and list next-day blockers

Milestones

Date	Milestone
22 Sep	Foundation + authentication + core CRUD + availability
24 Sep	Working appointment lifecycle
25 Sep	Queue system working
26 Sep	Smart slot recommendation working
27 Sep	Waiting-time + utilisation dashboard working
28 Sep	Audit + cancellation recovery working
29 Sep	Integrated system tested
30 Sep	Feature freeze + final demo/report

If You Fall Behind

Cut optional features first. The order to protect is: appointment booking → availability validation → queue → authentication/roles → dashboard → smart slot recommendation → waiting-time/utilisation → audit. The what-if simulator, notifications, attendance indicator and PDF export should be dropped before compromising the core system.

Definition of Done

By 30 September, a user should be able to log in, register/select a patient, manage doctors and availability, book/cancel/reschedule an appointment without conflicts, obtain a queue position, see an estimated wait, view operational dashboard information, and leave an auditable record of important changes. The standout smart-slot recommendation should also work end-to-end if included in the core build.